APU CLOUD INNOVATION HACKATHON 2026 · THEME 2



# RedPen

## An AI red team for university assessments

**They catch students.**
**We fix assignments.**

Muhammad Hassan · Muhammad Abdullah

THE PROBLEM

# Generative AI can now pass many university assignments.
# Lecturers find out at marking time.

### Too late

Exposure shows up after students submit, when the only tool left is an unreliable AI detector.

### Reported, not measured

"The MCCA has received a great deal of direct feedback from lecturers reporting a sharp increase in students submitting assignments almost entirely generated by AI."

### Never finished

Every stronger model makes last semester's "safe" assessment unsafe again. Redesign today is manual and one-off.

MCCA (Malaysia Cyber Consumer Association, an advocacy group), quoted by Bernama, 26 Jun 2025. Lecturer feedback, not survey data.

WHY IT MATTERS TO APU

# An assessment AI can pass no longer proves its learning outcomes.

- MQA requires assessment aligned to learning outcomes, with **CLO → PLO** mapping (COPPA 2nd ed.).
- APU's QAA accreditation is valid for five years, "subject to a satisfactory mid-cycle review". APU says that review **will take place by 2027**.
- So weak CLO evidence is a **university-level risk**, not one lecturer's headache.

**2027**

QAA mid-cycle review: the clock RedPen is built for.

MQA Code of Practice for Programme Accreditation (COPPA), 2nd ed. · qaa.ac.uk (APU accreditation, May 2024) · apu.edu.my

WHY NOW

# Leading universities are redesigning assessment. By hand.

### University of Sydney

Two-lane approach (secure and open assessment) from Semester 2 2025.

### University of Bath

"Closed" and "open" lanes replace its old categories from September 2026.

### AI Assessment Scale

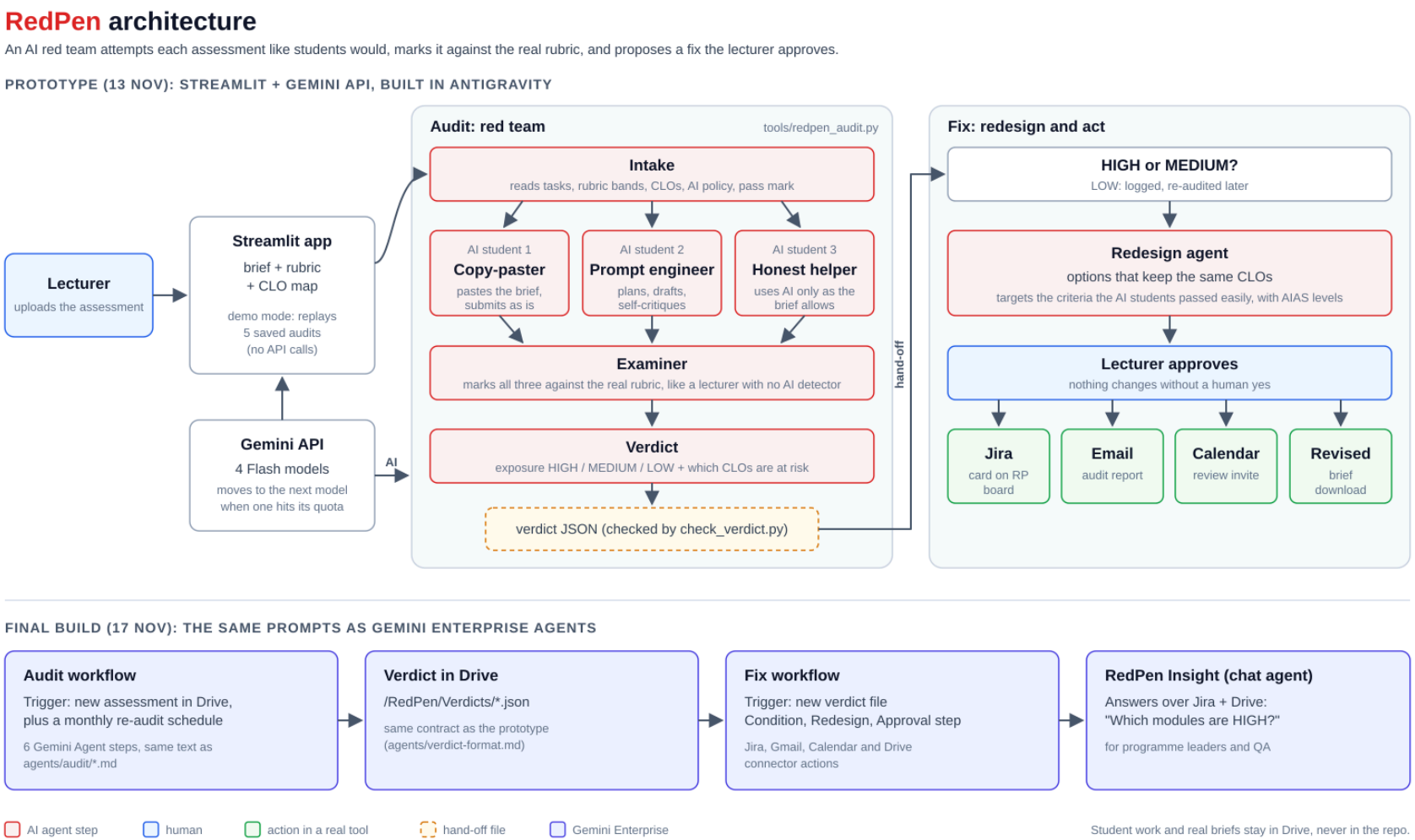
Five levels from No AI to AI Exploration (Perkins, Roe & Furze, 2025).

**The frameworks exist. The work of applying them, assessment by assessment, every semester, does not scale.** RedPen does that work.

sydney.edu.au (27 Nov 2024) · teachinghub.bath.ac.uk · Perkins, Roe & Furze (2025), JUTLP 22(7)

THE SOLUTION

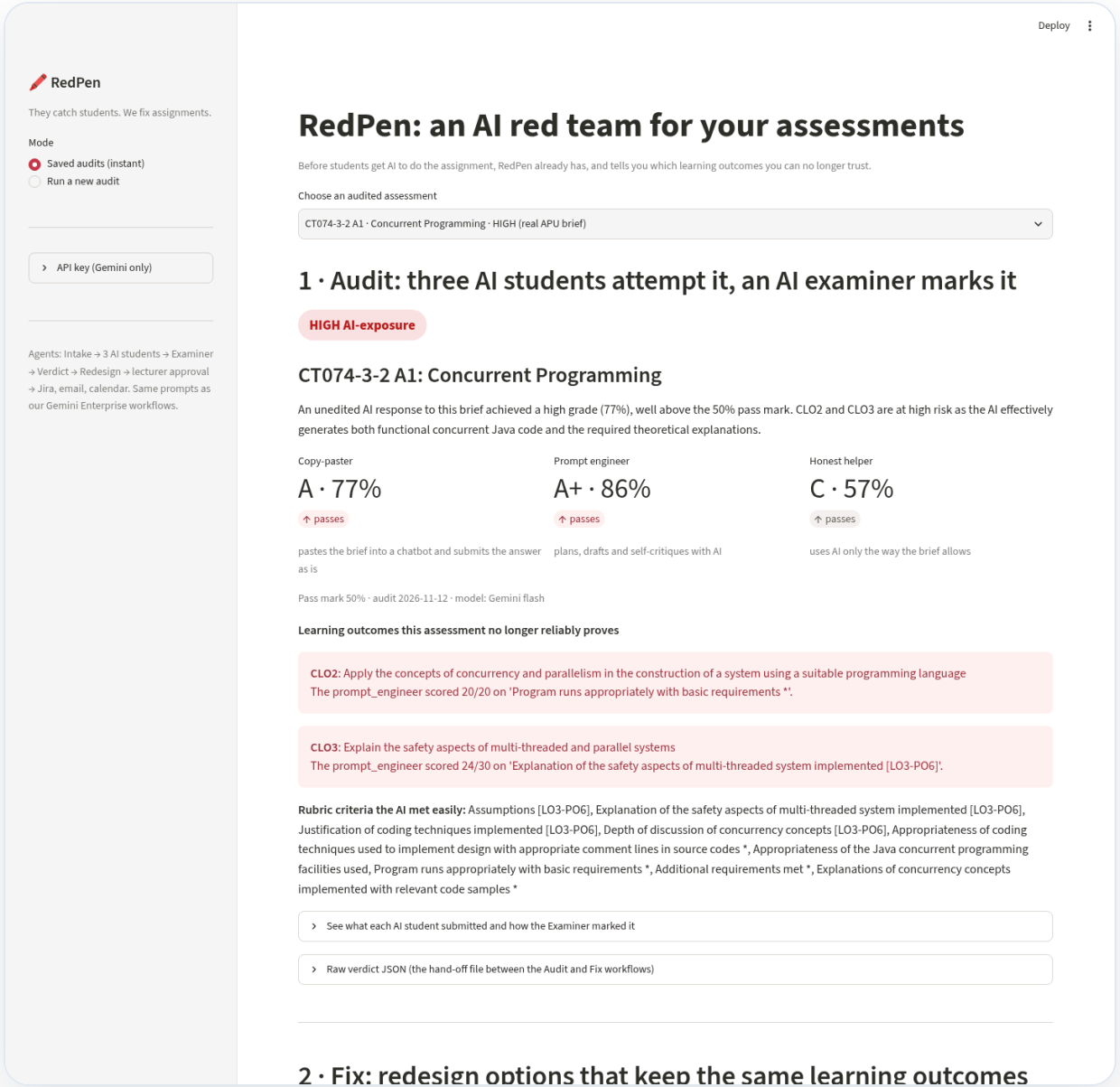
# Attack the assignment first. Then fix it, with the lecturer.



DEMO · REAL APU BRIEF

# Concurrent Programming: a copy-paste scores an A.

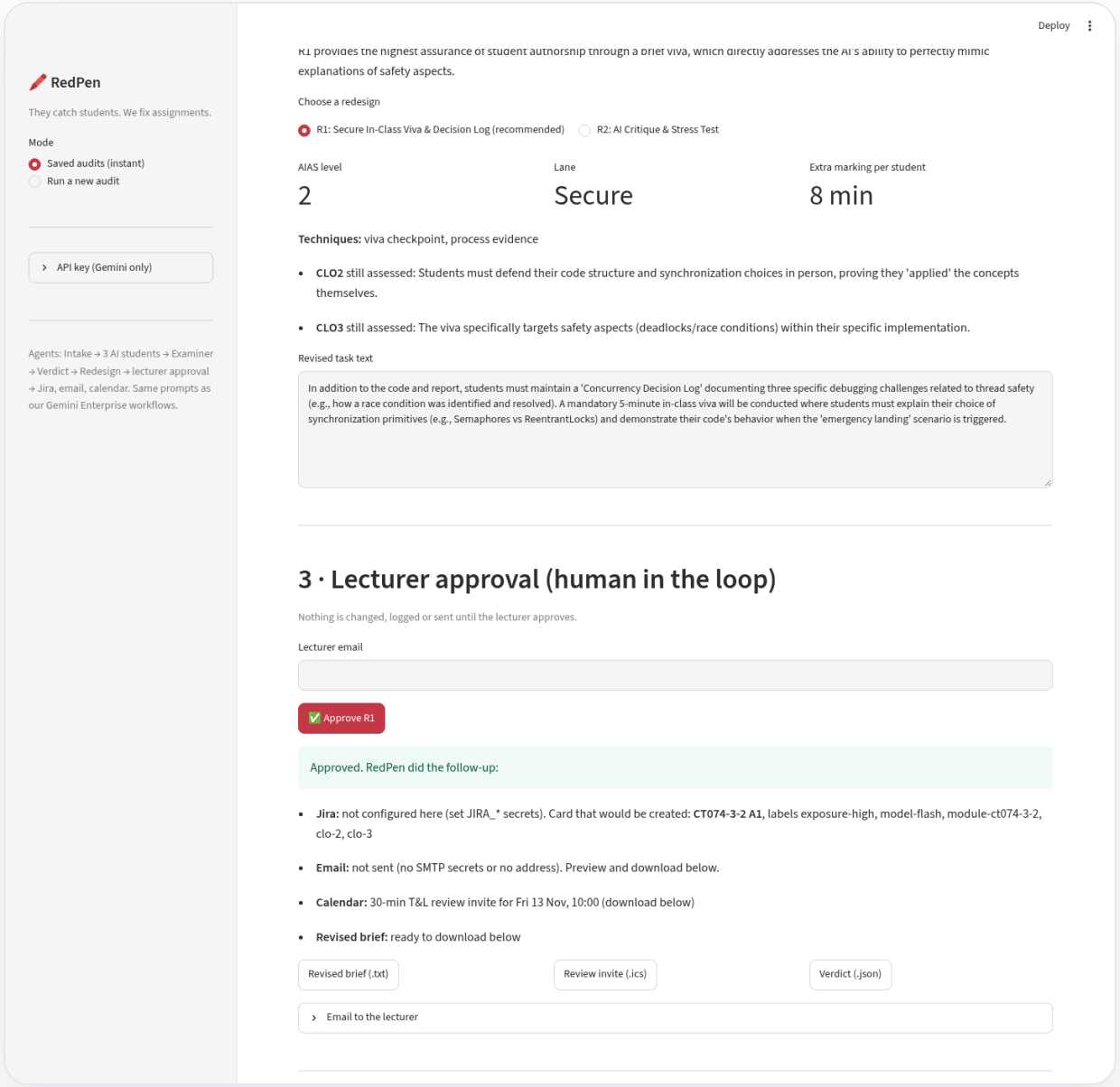
- **3 AI students** attempt it: a copy-paster, a prompt engineer, an honest helper.
- An AI **Examiner** marks them on the real rubric.
- Verdict: HIGH and **CLO2 + CLO3** no longer reliably proven.



DEMO · FIX + APPROVE

# Same learning outcomes. Harder to fake.

- 2–3 redesigns, each with an **AIAS level**, a lane and the extra marking time.
- Here: a decision log + 5-min in-class viva, **+8 min** marking per student.
- **Nothing happens until the lecturer approves.** Then: Jira register card, report email, review invite, revised brief.



PILOT · 4 REAL APU BRIEFS, THIS SEMESTER

# 2 of 4

passed by pasting the brief straight into Gemini.
**4 CLOs across 2 modules** no longer reliably evidenced.

| ASSESSMENT | COPY-PASTER | PROMPT ENG. | VERDICT |
|---|---|---|---|
| Concurrent Programming (CT074) | **A · 77%** | A+ · 86% | HIGH |
| Data Structures Lab 1 (CT077) | **C · 50%** | C · 57% | HIGH |
| Data Structures Lab 2 (CT077) | D · 47% | **A+ written part** | MEDIUM |
| Web Applications (CT050) | F · 33% | F · 37% | LOW |

Web Applications is LOW because half its marks come from a live demonstration to the lecturer: exactly the kind of design RedPen recommends. Pass mark 50% throughout. Gemini 3 Flash; full runs in the repo.

TRUST · "ISN'T THIS JUST AI GRADING AI?"

# 7 / 7

real APU assignments marked by our Examiner **within one grade band** of independent reference marks. 5 of 7 the same grade. Mean gap **4.4 points**.

- Reference marks were made by hand **before** any Gemini run, by a different model family. Not lecturer marks; we say so.
- Calibration caught a real flaw: the Examiner gave marks for a **viva that never happened**. Fixed: live-session criteria now score 0.
- The model matters: Flash and Pro gave different verdicts on 2 of 4 briefs. Every verdict records its model, and RedPen **re-audits monthly**.
- A lecturer approves every change. RedPen never judges a real student.

HOW IT'S BUILT

# One set of agents. Prototype today, Gemini Enterprise ready.

### Prototype (today, on Google Gemini)

8 agents: Intake, 3 AI students, Examiner, Verdict, Redesign, **Insight chat agent**. Lecturer approval → Jira register, email, calendar invite, revised brief. A full audit takes **~30 s**.

### Gemini Enterprise (built for it, not ported later)

**Audit workflow** (Drive trigger + monthly schedule), **Fix workflow** (approval step + Jira, Gmail, Calendar, Drive actions), **Insight** chat agent over Jira + Drive.

**Same prompts, generated** paste sheet: what we tested is exactly what we paste.

**Worst case tested:** steps that only see the previous step still produce valid verdicts.

**32 automated tests**, a JSON schema per step, and a checker for every Gemini Enterprise run.

**Google Antigravity:** [fill in: what we used it for]

IMPACT

# Minutes per assessment, not a semester of committee work.

**<1 min**

per full audit on Gemini Flash (6 agent steps); 3–6 min on Pro.

**3 mo**

One programme's whole assessment bank audited and fixed before the semester starts.

**6 mo**

Every programme, re-audited monthly, with a trend for the QAA mid-cycle review.

Fewer misconduct cases to argue over. Stronger CLO evidence. Reusable by any university.

WHY REDPEN · WHO BUILT IT

# Detection polices students after they submit. RedPen prevents, before release.

Complementary to detection tools, but prevention is cheaper, less adversarial, and it protects the evidence the university reports to regulators.

| WHO | BUILT |
|---|---|
| **Abdullah** | Audit agents (intake, 3 AI students, examiner, verdict), calibration, pilot runs, prototype app, architecture, video |
| **Hassan** | Fix agents (redesign, approval, Jira, email, calendar), Jira site, research and fact-checks, lecturer outreach, deck |

One shared contract between the two halves: the verdict JSON.
Repo: github.com/L0adn9/redPen

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